

# ALEX YAN

U.S. Citizen | Wexford, PA | 412-699-3595 | [alexyan2309@gmail.com](mailto:alexyan2309@gmail.com) | [Linkedin](#) | [Github](#) | [Personal Website](#)

## EDUCATION

### Purdue University

*B.S. Computer Science*

West Lafayette, IN

May 2028

- **Relevant Coursework:** Data Structures & Algorithms, Object-Oriented Programming, Systems Programming, Computer Architecture, Programming in C, Discrete Mathematics, Linear Algebra, Introduction to Statistics

## EXPERIENCE

### Shopify

*Software Engineer Intern*

New York, NY

Sep. 2026 – Present

- Build Ruby on Rails and GraphQL backend services for Shopify Payments' bank-account ownership verification, which confirms a merchant's payout account belongs to them before funds are released.
- Cut third-party verification costs by fixing a retry bug that created a new paid identity check per retry; used idempotent request IDs and lookup-before-create recovery, with no schema migration.
- Made multi-step verification updates atomic with database transactions, and hardened sensitive-data handling with sanitized error paths and a reusable masking module that keeps account details out of logs.
- Added ownership-verification status visibility for Shopify's support team in an internal tool, tested end to end against a live local stack.

### PNC Bank

*Software Engineer Intern*

Pittsburgh, PA

Jun. 2026 – Aug. 2026

- Delivered Spring Boot validation features in Java for loan origination workflows on the application entry REST API, deployed through Red Hat OpenShift.
- Expanded JUnit coverage for validation logic and verified edge cases end-to-end with Postman against MongoDB-backed configuration; fixed intermittent Jenkins CI failures caused by race conditions in asynchronous tests.
- Built an LLM agent that catches convention and coverage gaps in unit tests before PR submission, grounded in team testing patterns and design discussion to cut PRs needed for approval by 55%.

### Purdue Stack

*Full Stack Developer*

West Lafayette, IN

Sep. 2025 – May 2026

- Shipped a research collaboration platform in Next.js, FastAPI, and PostgreSQL for Purdue's Statistics Department, featuring role-based dashboards, real-time messaging, and JWT authentication. (Spring 2026)
- Cut TA scheduling setup time 25% with a preference-weighted greedy matching algorithm in Python, powering an assignment system for 150–200 TAs per semester in Purdue's Mathematics Department. (Fall 2025)

### Rooh (Berkeley SkyDeck Pad-13 Incubator)

*Software Engineer Intern*

Berkeley, CA (Remote)

Jun. 2025 – Dec. 2025

- Cut voice response latency 20% by rearchitecting an end-to-end Google Cloud STT → LLM → TTS pipeline for a real-time conversational AI app in React and Next.js.
- Architected Next.js backend data pipelines supporting a live AI simulator and async review tool, handling real-time audio streaming, buffering, and transcription at scale.

## PROJECTS

### Raft Consensus Implementation (MIT 6.5840) | *Go, RPC, Concurrency* | [Github](#)

Oct. 2026

- Implemented the Raft consensus algorithm in Go: randomized-timeout leader election, log replication across peers, and crash-durable state, keeping replicas' state machines identical under network partitions, dropped messages, and server restarts.
- Handled concurrent RPC state transitions with per-peer replication tracking, term-guarded reply handling, and term-aware log backtracking to repair divergent follower logs, passing the full test suite under Go's race detector.

### Perch | *Electron, TypeScript, React, SQLite, OpenAI API, WebSockets* | [Github](#)

Jul. 2026

- Built a local-first desktop focus app that generates a personalized AI mascot from a user's photo and delivers real-time, task-aware nudges on detecting off-task activity, tracking native window state and browser events through a paired Chrome extension.
- Designed a tiered classification pipeline escalating ambiguous states to OpenAI vision, with structured output validation, static fallbacks, and non-repeating message pools for reliable UX under API failure.

### Access Prep | *React, TypeScript, Flask, LangChain, Gemini* | [Github](#)

Jan. 2026

- **3rd Place** at the Capital One Hackathon; document prep platform with a Flask REST API and LangChain/Gemini guidance chatbot.

## SKILLS

**Languages:** Python, Java, TypeScript, Go, SQL, JavaScript, C++, Ruby, C, Bash

**Frameworks & Libraries:** Spring Boot, React, Ruby on Rails, Node.js, Next.js, FastAPI, Flask

**Systems & Protocols:** REST APIs, Distributed Systems, Concurrency, GraphQL, RPC, CI/CD

**Databases & Infrastructure:** PostgreSQL, AWS (Lambda, API Gateway), Docker, MongoDB, Google Cloud, SQLite, OpenShift

**Tools & Testing:** Git, JUnit, GitHub Actions, Jenkins, Postman, Claude Code, GitHub Copilot, Cursor